

Skills Summary

Solving Quadratics Without Losing a Root or a Sign

Use this reference while completing your worksheet or when studying.

Skill 1 — Every square root gets a plus-or-minus

Rule: if $u^2 = k$ and $k > 0$, then $u = \pm\sqrt{k}$ — two numbers share one square, so one square root step produces *two* branches.

- Never divide both sides by something containing x : that erases the root where it equals zero. Factor instead.
- How many roots to expect: discriminant positive $\rightarrow$ two, zero $\rightarrow$ one (a double root), negative $\rightarrow$ none real.
- Finish by substituting every root back into the original equation.

Example: Solve $x^2 = 36$.

Step 1. The variable appears only as a square, so this is a square-root problem — and the square-root step is exactly where a root goes missing.

Step 2. $x = \pm\sqrt{36}$, which splits into the two branches $x = +6$ and $x = -6$.
 $x = -6$ or $x = 6$

Try it: Solve $x^2 = 81$. Give the complete solution set.

check: $6 = x$ or $6 = -x$

Skill 2 — Substitute signs, not just digits

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, $\Delta = b^2 - 4ac$.

Two sign traps. $-b$ means *the opposite of b* : if b is negative, $-b$ is positive. And $-4ac$ becomes an *addition* whenever a and c have opposite signs. Write a , b , c with their signs in a list before substituting, and put every negative in brackets.

Example: Solve $x^2 - 6x + 5 = 0$ with the formula; give the larger root.

Step 1. Nothing is squared alone here, so the formula is the general tool — and b is negative, which is where the sign slip happens.

Step 2. $a = 1$, $b = -6$, $c = 5$, so $-b = 6$ and $\sqrt{b^2 - 4ac} = 4$: $x = \frac{6 + 4}{2}$.

The larger root is $x = 5$.

Try it: Solve $x^2 - 2x - 8 = 0$ with the formula and give the larger root.

check: $4 = x$

Skill 3 — Reading a factored equation correctly

Zero-product property: if $PQ = 0$ then $P = 0$ or $Q = 0$. Set *each factor* equal to zero and solve it as its own little equation.

- The root has the *opposite* sign to the constant in its factor.
- A factor with a coefficient needs two moves, not one: subtract, then divide.
- Count your roots: one factor, one root.

Example: Solve $(3x - 2)(x + 5) = 0$.

Step 1. The product is already zero and already factored, so the zero-product property finishes it — no expanding.

Step 2. $3x - 2 = 0$ needs two moves: $3x = 2$, then divide by 3. And $x + 5 = 0$ gives $x = -5$.
 $x = \frac{2}{3}$ or $x = -5$

Try it: Solve $(2x - 5)(x + 1) = 0$.

$1 - = x \text{ to } \frac{2}{5} = x$:xæp

(!) Watch out: A solution set with one root is a claim, not a default. Either the discriminant is zero, or you lost a root at a square root or a division. Check which before you hand it in.
